Jiaqi Wu

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Education

University of California San Diego Expected Dec 2026
B.S. in Mathematics–Computer Science
B.S. in Cognitive Science, Specialization in Machine Learning and Neural Computation
GPA: 3.967/4.00

Research Experience

Closed-Loop Self-Correcting Image Editing Apr 2026 – Present
Research Assistant Intern, mlPC Lab, UC San Diego | Advisor: Prof. Zhuowen Tu

- Designing a closed-loop framework that uses feedback control to iteratively correct a frozen diffusion editor.
- Developing alternating on-policy training for correction and execution modules; evaluating edit accuracy and content preservation against equal-compute best-of-N resampling.
- Constructing synthetic source-image, edit-instruction, and target-image triplets from video-model outputs, filtered using SAM, CLIP, and VLM scoring.

BrainMech: Modeling Human Brain Dynamics Sep 2025 – Jul 2026
Research Assistant Intern, ACM Lab, UNC Chapel Hill | Advisor: Prof. Guorong Wu

- Proposed a self-supervised pretraining objective combining next-step forecasting, latent dynamics consistency, and spectral stability regularization for a Koopman-based fMRI nonlinear dynamics foundation model.
- Proposed functional connectivity preprocessing procedures and contributed to pretraining on resting-state fMRI cohorts spanning development and aging.
- Proposed region-level mediation analyses to examine whether learned representations mediate associations between regional BOLD signals and disease labels.

Preprints & Manuscripts

[1] Shuheng Cao*, Weijia Zhang*, **Jiaqi Wu***, Xiyun Hu, Yat Yang, Juqy Chen, and Zhaoxiang Feng. *KNOWPLAN: Knowledge-Driven AI Agents for Smart Degree Pathway Planning*. arXiv:2608.06530, 2026. *Co-first authors.

[2] **Jiaqi Wu**, Shuheng Cao, Tingting Dan, Ziquan Wei, Zhixuan Zhou, and Guorong Wu. *A Mechanistic Foundation Model for Human Brain Dynamics Across Lifespan*. Unpublished manuscript.

Selected Project

KnowPlan (Academic Planning Software) Preprint: 2026
Co-first author | LLM integration and data collection

- Integrated LLM components and collected university curriculum data for KnowPlan, a framework combining curriculum extraction with constraint-based degree planning.
- In related course-planning work, built a multi-agent GraphRAG pipeline, SwiftUI app, and AWS serverless backend for prerequisite-aware retrieval and multi-term scheduling.
- Developing academic planning software toward public release.

Teaching & Recognition

MATLAB Tutor, Department of Mathematics Mar 2025 – Dec 2025
UC San Diego

- Provided MATLAB tutoring for linear algebra and differential equations courses with 1,000+ combined enrollments, explaining concepts, debugging code, and guiding students through programming assignments.

Award: SanD Hackathon — Multi-Agent System Challenge Winner, San Diego

Technical Skills

Programming: Python, C/C++, MATLAB, JavaScript, Swift
Machine Learning: PyTorch, Hugging Face Transformers, Accelerate, FSDP, NumPy, pandas, scikit-learn
Tools: Git, Linux, AWS, LATEX